



Asymptotic decay rate of an infinite product of sinc functions

Asked 8 years, 4 months ago Modified 7 years, 7 months ago Viewed 1k times



(related to my [previous question](#))

20

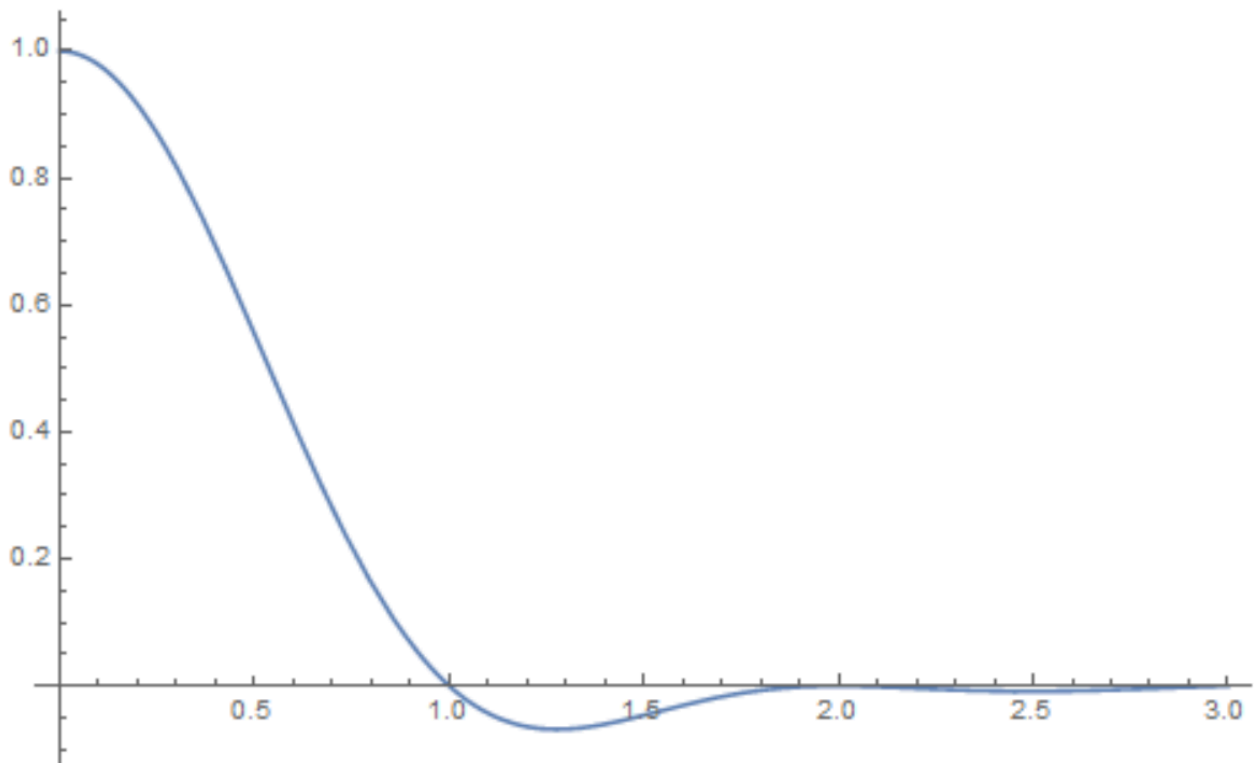
Consider the function



$$f(x) = \prod_{n=0}^{\infty} \operatorname{sinc}\left(\frac{\pi x}{2^n}\right), \quad x \geq 0, \quad (1)$$



where $\operatorname{sinc}(z)$ denotes the [sinc function](#). The function $f(x)$ has zeros at positive integers, and oscillates with a quickly decaying amplitude. Its signs on the intervals between consecutive zeros follow the same pattern as the [Thue–Morse sequence](#).



Can we find a real-valued function $g(x)$ (elementary, if possible), such that it is analytic, positive and monotone decreasing (and having monotone derivatives of any order, if possible) and satisfies

$$\lim_{t \rightarrow \infty} \frac{1}{t - t_0} \int_{t_0}^t \frac{|f(x)|}{g(x)} dx = 1 \quad (2)$$

for large enough t_0 ? Or, at least,

$$\exists A > 0, \exists B > A, \forall t > t_0, A < \frac{1}{t - t_0} \int_{t_0}^t \frac{|f(x)|}{g(x)} dx < B \quad (3)$$

for large enough t_0 ?

The "*for large enough*" provision means that $g(x)$ is permitted to be not monotone, or have zeros or discontinuities, or be not defined at all for small t , and we only care about its "eventual" behavior.

Empirically, it looks like $g(x)$ should decay faster than any negative power of x , but slower than exponentially. I'm thinking something close to $\exp(-\log^2 x)$, but perhaps not exactly that.

Update: Maybe this expansion can be useful:

$$\prod_{n=0}^m \operatorname{sinc}\left(\frac{\pi x}{2^n}\right) = \frac{2^{\binom{m}{2}}}{(\pi x)^{m+1}} \sum_{n=0}^{2^m-1} t_n \sin\left(\frac{\pi m}{2} + \frac{2n+1}{2^m} \pi x\right), \quad (4)$$

where $t_0 = 1$, $t_n = (-1)^n t_{\lfloor n/2 \rfloor}$ (the signed Thue–Morse sequence).

[calculus](#) [sequences-and-series](#) [limits](#) [asymptotics](#) [infinite-product](#) [Edit tags](#)

[Share](#) [Cite](#) [Edit](#) [Close](#) [Delete](#) [Flag](#)

edited Apr 21, 2018 at 20:48

asked Apr 20, 2018 at 3:24



Vladimir Reshetnikov

33.5k 12 185 331



Don't you have the limit in (2) equal to 0 for t_0 an integer? – [zhw](#). May 1, 2018 at 0:33 [✎](#)



@zhw. A choice of t_0 does not affect the limit (if it exists), because we are averaging the ratio of functions over an infinite ray. Changing something in a finite initial segment of it does not affect the result. – [Vladimir Reshetnikov](#) May 1, 2018 at 22:58



Ah sorry, my bad. Didn't see $t \rightarrow \infty$ – [zhw](#). May 2, 2018 at 0:08 [✎](#)



1 Answer

Sorted by: [Highest score \(default\)](#) [⬆](#)



3

Definition



Let



+300

Formulas for $\text{sinc}(\pi z)$ 

Using symmetry formulas for the gamma function

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)} \quad (2)$$

easy to obtain

$$\frac{\sin(\pi z)}{\pi z} = \frac{1}{z\Gamma(z)\Gamma(1-z)},$$

$$\text{sinc}(\pi z) = \frac{1}{\Gamma(1+z)\Gamma(1-z)}, \quad (3)$$

$$\ln \text{sinc}(\pi z) = -(\ln \Gamma(1+z) + \ln \Gamma(1-z)). \quad (4)$$

Using the Maclaurin series in the form of

$$\ln \Gamma(1+z) = -\ln(1+z) + (1-\gamma)z + \sum_{j=2}^{\infty} (-1)^j \frac{\zeta(j)-1}{j} z^j, \quad |z| < 2, \quad (5)$$

where $\gamma = 0.57721\dots$, one can get

$$\ln \Gamma(1-z) = -\ln(1-z) - (1-\gamma)z + \sum_{j=2}^{\infty} \frac{\zeta(j)-1}{j} z^j,$$

$$\ln \text{sinc}(\pi z) = \ln(1-z^2) - \sum_{k=1}^{\infty} \frac{\zeta(2k)-1}{k} z^{2k}.$$

Zeta function of the even argument is

$$\zeta(2k) = \frac{(2\pi)^{2k}}{2(2k)!} |B_{2k}|, \quad (6)$$

where B_{2k} are the [Bernoulli numbers](#). Then

$$\ln \text{sinc}(\pi z) = \ln(1-z^2) - \sum_{k=1}^{\infty} c_k z^{2k}, \quad (7)$$

where

$$c_k = \frac{1}{2k} \left(\frac{(2\pi)^{2k}}{(2k)!} |B_{2k}| - 2 \right). \quad (8)$$

Formulas for $R(x)$

Let us find the sum

$$\begin{aligned} \ln R_m(x) &= \sum_{n=m}^{\infty} \ln \operatorname{sinc} \frac{\pi x}{2^n}, \\ \ln R_m(x) &= \sum_{n=m}^{\infty} \left(\ln \left(1 - \left(\frac{x}{2^n} \right)^2 \right) - \sum_{k=1}^{\infty} c_k \left(\frac{x}{2^n} \right)^{2k} \right) = \\ &= \sum_{n=m}^{\infty} \ln \left(1 - \left(\frac{x}{2^n} \right)^2 \right) - \sum_{k=1}^{\infty} \left(c_k x^{2k} \sum_{n=m}^{\infty} 2^{-2nk} \right) = \\ &= \sum_{n=m}^{\infty} \ln \left(1 - \left(\frac{x}{2^n} \right)^2 \right) - \sum_{k=1}^{\infty} \frac{c_k}{1 - 2^{-2k}} \left(\frac{x}{2^m} \right)^{2k}. \end{aligned}$$

The first term can be presented over the [q-Pochhammer Symbol](#) in the form of

$$\sum_{n=0}^m \ln \left(1 - \left(\frac{x}{2^n} \right)^2 \right) = \ln \left(x^2, \frac{1}{4} \right)_m. \quad (9)$$

This allows to obtain the formulas

$$R(x) = \prod_{n=0}^{m-1} \operatorname{sinc} \frac{\pi x}{2^n} \cdot \frac{\left(x^2, \frac{1}{4} \right)_{\infty}}{\left(x^2, \frac{1}{4} \right)_m} \cdot \exp \left(\sum_{k=1}^{\infty} \frac{c_k}{1 - 2^{-mk}} \left(\frac{x}{2^m} \right)^{2k} \right), \quad |x| < 2^{m+1}. \quad (10)$$

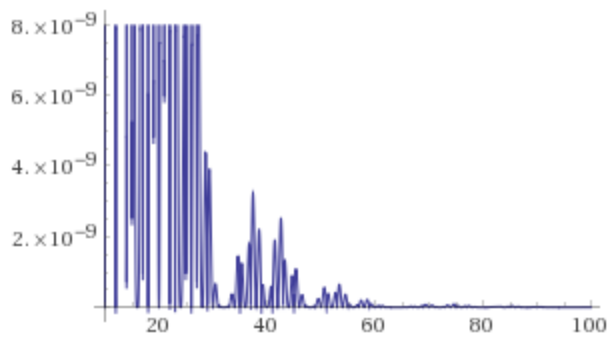
Radius of convergence in the formula (10) shows that it gives $R(x) = 0$ outside this radius, such as the Stirling asymptotics. Easy to see that parameter m allows to manage this radius. At the same time, it allows to provide the required length of sum.

If the bounds of the required interval are predefined, then parameter m can be considered as constant. This allows to use the common function for this interval.

For example, for the predefined interval $x = (10, 100)$ the value $m = 6$ can be chosen. Then formula (10) presents the function $|R(x)|$ as the production of the three factors:

[partial sinc production](#),

Plot:

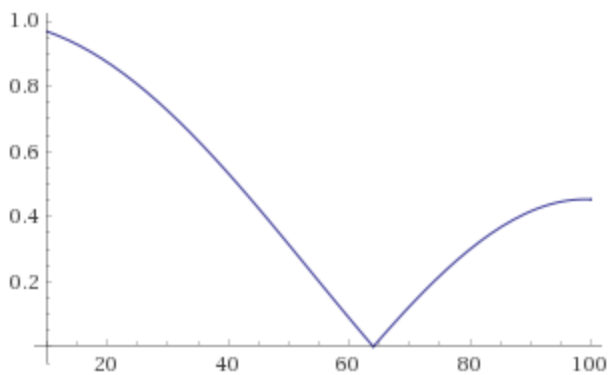


plot prod_{n = 0}⁵ | ((pi * x * 2⁽⁻ⁿ⁾) sin(pi * x * 2⁽⁻ⁿ⁾) /
((pi * x * 2⁽⁻ⁿ⁾)² + 10⁽⁻¹⁰⁾) |,
where x in (10, 100)

Computed by Wolfram|Alpha

q-Pochhammer ratio

Plot:

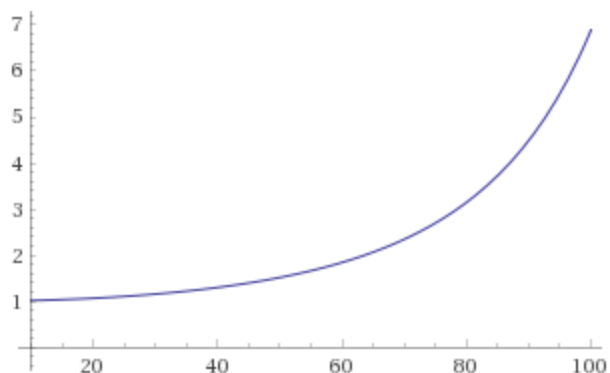


plot (prod_{n = 5}²⁰ (1.0 - x² 2^(-2 * n - 2))) *
sign(64 - x), where x in (10, 100)

Computed by Wolfram|Alpha

and the reminder series exponent

Plot:



plot prod_{n =}
1³ e<sup>((| bernoulli(2 n) | / (2 n)! (2 pi)^(2 n - 2) /
(1 - 64⁽⁻ⁿ⁾))
(x / 64)^{(2 n) / (2 n)}), where x = 10, 100</sup>

Computed by Wolfram|Alpha

In the general case, the most convenient way looks using of formula (10) for a piecewise approximation on the required intervals .

Anyway, this allows to take in account the multiplicity of preceeding zeros.

Share Cite Edit Follow Flag

edited Jan 24, 2019 at 19:52

answered Apr 30, 2018 at 0:29



Yuri Negometyanov

29.3k 1 36 64

Anyway, the function $f(x)$ is known to be continuous and analytic, its Taylor–Maclaurin series and Fourier transform are studied pretty well, see, for example, the references in my [previous question](#).

– Vladimir Reshetnikov Apr 30, 2018 at 3:56



@VladimirReshetnikov New version. Waiting for comments. – Yuri Negometyanov May 1, 2018 at 5:06

In formula (10), there is a free variable m on the right, but not on the left -- is it a typo? Also, the summation over k in (10) starts from $k = 0$, but the summand contains c_k . Looking at the definition (8), we can see that c_0 is undefined because it involves a division by zero. Also, there is another division by zero because of the denominator $1 - 2^{-mk}$. – Vladimir Reshetnikov May 1, 2018 at 23:10



@VladimirReshetnikov Of course, this was a typo. The earlier formulas contains $k = 1$. Thank you for accuracy. – Yuri Negometyanov May 1, 2018 at 23:20



@VladimirReshetnikov By the way. Is known the inequality

$$\frac{2(2n)!}{(2\pi)^{2n}} \left(\frac{1}{1 - 2^{1-2n}} \right) > (-1)^{n+1} B_{2n} > \frac{2(2n)!}{(2\pi)^{2n}}, \quad n = 1, 2, \dots$$

– Yuri Negometyanov May 1, 2018 at 23:36

What about the free variable m in (10)? – Vladimir Reshetnikov May 2, 2018 at 0:26

I wonder where the scope of the $\prod$ in (10) ends. Is it over only the first factor, or all three?

– Vladimir Reshetnikov May 2, 2018 at 0:30



@VladimirReshetnikov Really, (10) contains the ratio of $R(x) = R_\infty(x)$ and $R_m(x)$, and that allows to acheve the nesessary (arbitrary) accuracy. Taking in account the convergency raduis and the required possibly closed form, it does not looking so free. – Yuri Negometyanov May 2, 2018 at 6:32



@VladimirReshetnikov It is possible to bound the required series length in (10). At the same time, it gives 0 if $x \geq 2^{m+1}$. And I don't see any way to get the asymptotic decay rate as the universal function in the closed form. – Yuri Negometyanov May 2, 2018 at 6:52

Thank you for the effort (and the bounty is yours), but I do not think this answer brings us significantly closer to resolving my question. – Vladimir Reshetnikov May 3, 2018 at 18:19



@VladimirReshetnikov The problem is solved in closed form for the 'local infinity' settings (see the updated text). I think this is the maximum of possible. – Yuri Negometyanov May 4, 2018 at 1:40